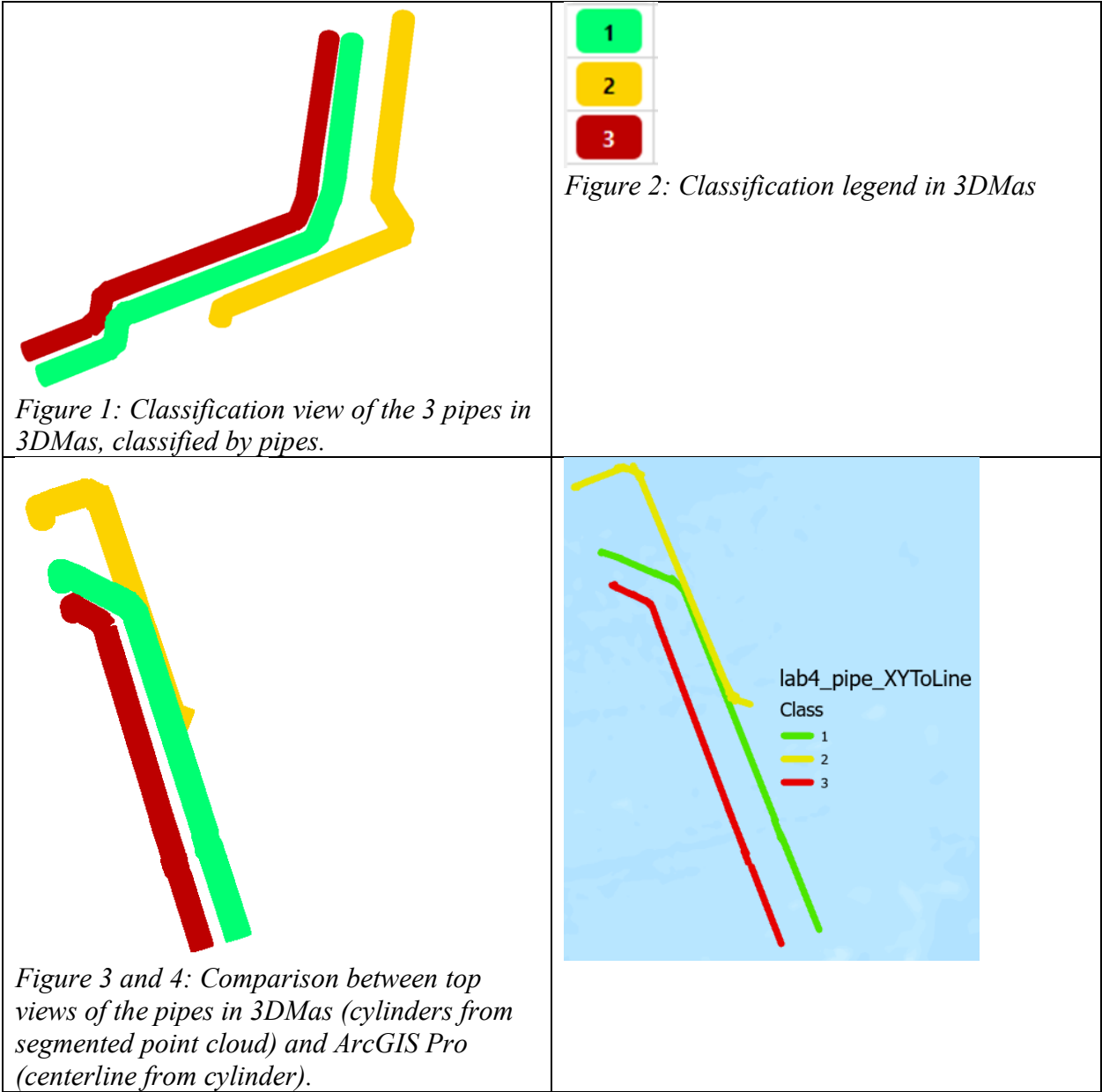


LSGI3262 Laser Scanning – Lab 4

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Student ID: [redacted]

Task 1 - Screenshot of cylinder-fitted data in 3DMas and centerline in ArcGIS Pro



Task 2 – Description of the information extraction process

Step 1: Slicing (Manual preprocessing)

The raw point cloud consists of all three pipes together. Although they do not appear to have overlapping regions, segmented pipes are often easier to work with the Polygon Selection Tool to manually select the points for precise cylindrical fitting. Only the points relevant to the pipe should be selected for cylindrical fitting, or else the fitting algorithm may inadvertently include points from adjacent pipes. Therefore, this is done so that only the points belonging to the target are captured without accidentally including the points from adjacent pipes.

To overcome these issues, such as accidentally including points from neighboring pipes when selecting points for fitting, the point cloud was manually segmented into individual pipes using Polygon Selection Tool in 3DMas.

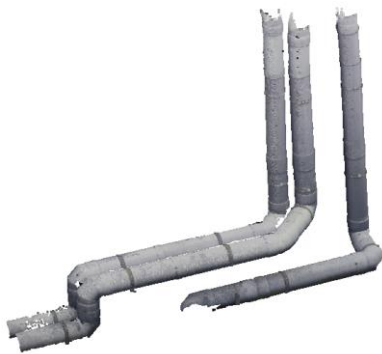


Figure 5: Raw point cloud data

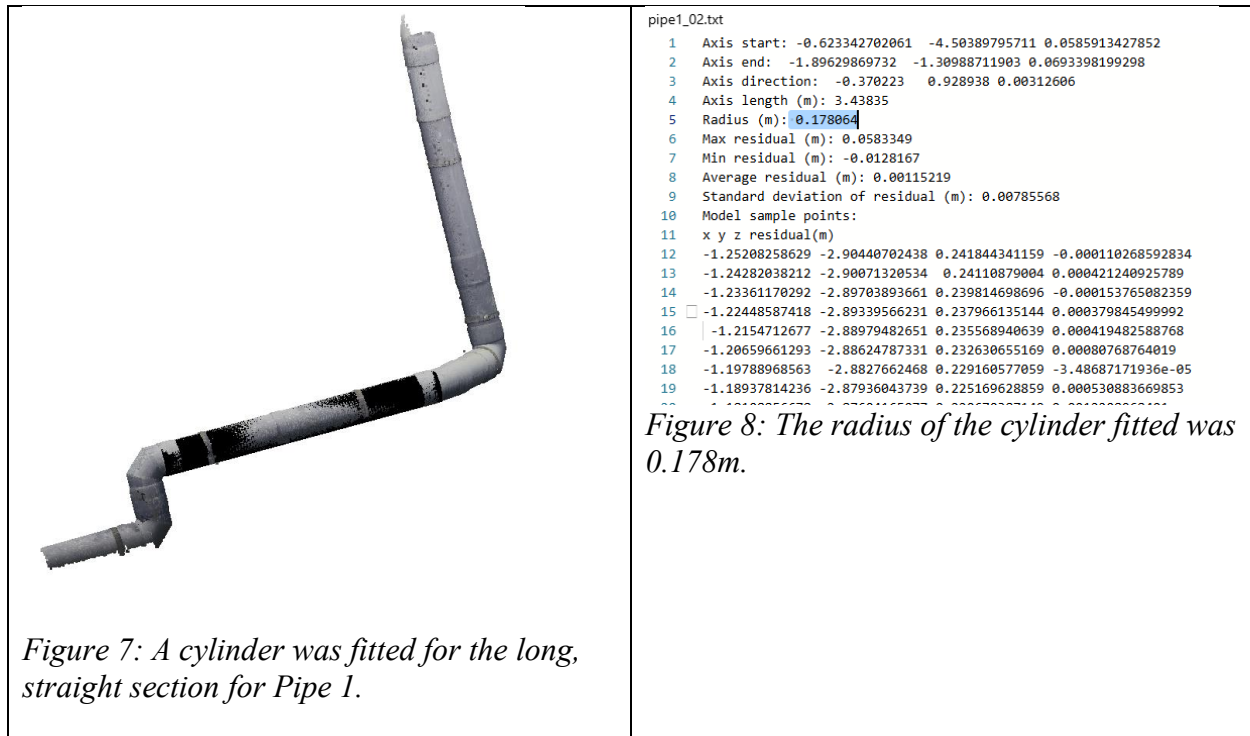


Figure 6: After slicing one pipe

After the separation, the points are saved as different classes, and can be hidden from view and locked to prevent further modification. The images show a comparison between the unsliced and sliced data, providing a clean and focused subset data (one single pipe at a time) for fitting.

Step 2: Adaptive radius cylindrical fitting for long straight sections (Semi-supervised fitting)

Long straight pipe sections were fitted using the Adaptive Radius option of the Cylinder Fitting function. Adaptive fitting was chosen for these sections because long, straight runs typically have high point density, consistent and uniform shape of the pipe structure that maintains the pipe radius, giving the algorithm a high probability of converging to a correct and accurate fit. Furthermore, it does not require the user to input an initial known or guessing radius, which can make the software more convenient.

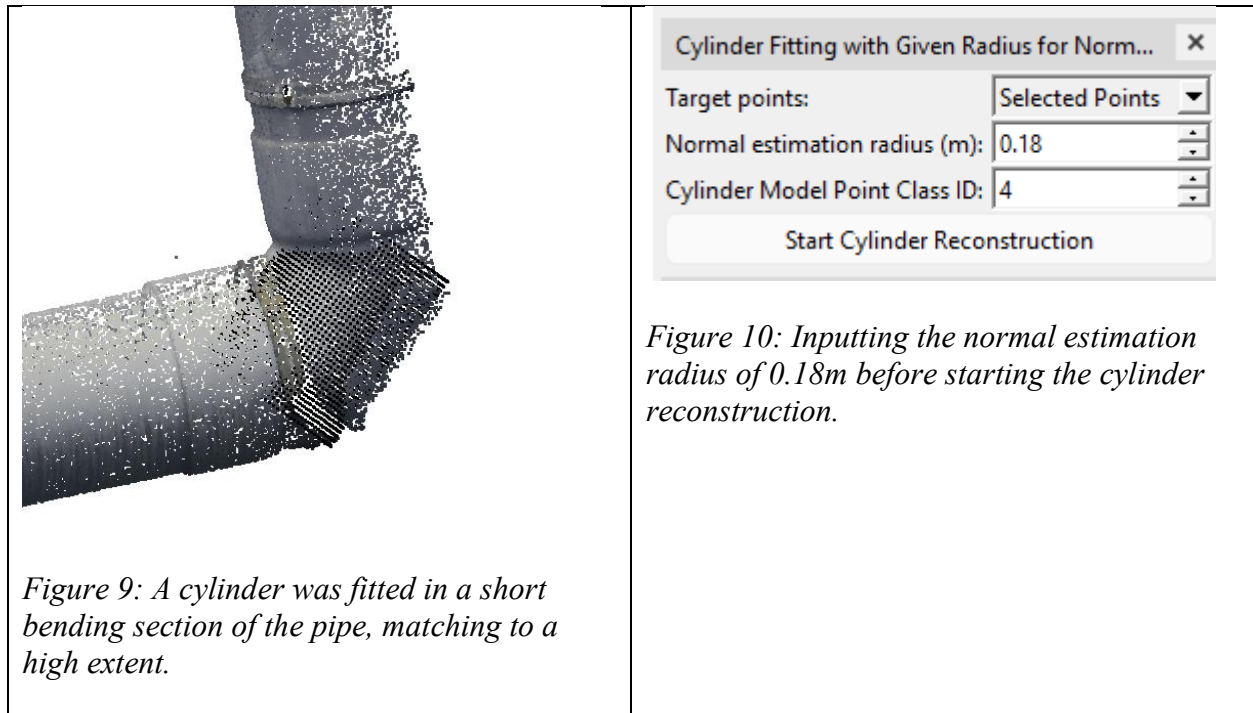


After multiple segments, we can view the pipe radius from the output text. The numbers showed on average 0.18m consistently, confirming a representative pipe radius, therefore we can take this value as a reliable reference radius for subsequent cylindrical fitting for other sections.

Step 3: Cylindrical fitting using a normal estimation radius (0.18m)

In the point cloud data of the pipes, there were some sections that are curved or bent, and using the adaptive cylindrical fitting might be inaccurate and computationally more expensive. To address this, a surface normal estimation radius was applied.

The fixed radius of 0.18 m was derived from the average obtained in Step 2, assuming the pipes maintain a constant nominal diameter. This constrained fitting reduces the degrees of freedom, leading to faster convergence and more stable results on short, bent or low-density sections compared to fully adaptive methods.



Cylinder Fitting with Given Radius for Norm... X

Target points: Selected Points

Normal estimation radius (m): 0.18

Cylinder Model Point Class ID: 4

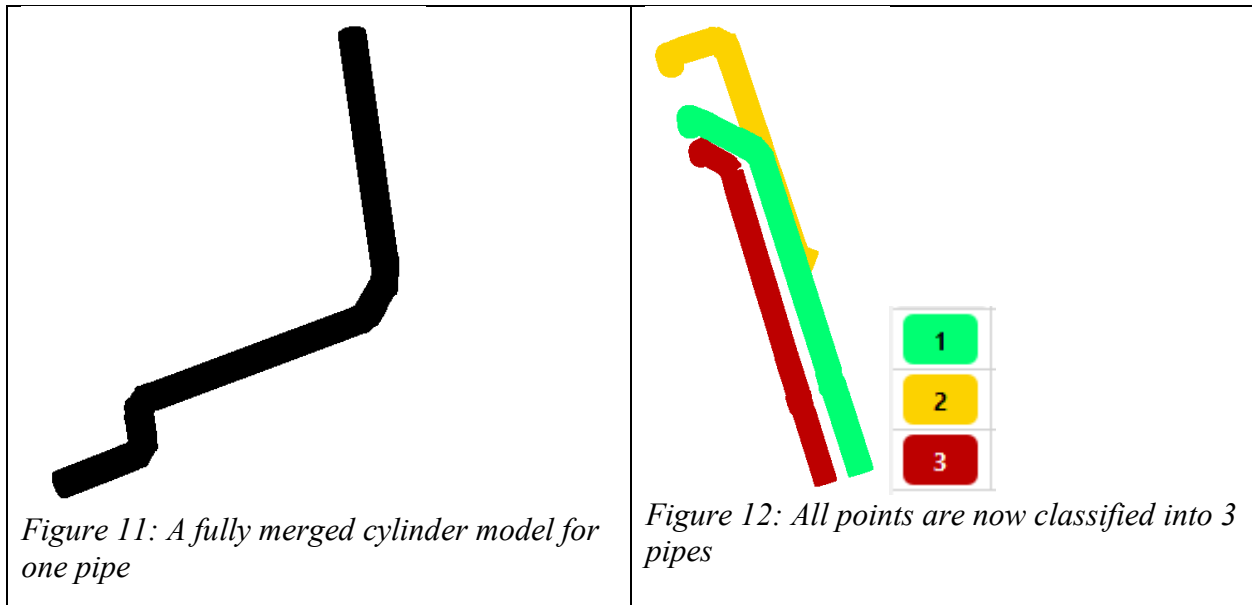
Start Cylinder Reconstruction

Figure 10: Inputting the normal estimation radius of 0.18m before starting the cylinder reconstruction.

From the figures, the visual inspections confirmed that this method produced smooth and geometrically consistent fits on the bends, with high degree of alignment to the original point clouds and generally faster convergence. This approach avoided common issues with adaptive fitting on elbows of pipes, such as radius being inconsistent with similar selected points.

Step 4: Merge the cylinders to their respective classes

After fitting the cylinders, the class ID might be different. This is to avoid directly merging with the pipes in case of undesirable fitting results. Under the user's supervision, the pipes can be grouped and merged according to their assigned pipe class. For Pipe 1 and 3, 15 cylinders were used for each class, while for Pipe 2, 13 cylinders were used.



Merging groups the individual segments, while creating the complete pipe models enables us to make further manual adjustments. For example, the pipes can be made smoother that matches the typical shape of the pipe, and remove the points from the cylinder fitting that extend beyond the pipe. Figure 11 shows the cylinders fully merged to form Pipe 1, demonstrating that refining the model after merging can achieve seamless connection of straight and bent sections.

Step 5: Produce a CSV with start and end point XYZ coordinates, use ArcGIS Pro XY to Line function to create line feature

During the cylindrical fitting process, the software saves the cylinder information to a text file, which contains the starting and ending coordinates of the cylinder. The fields were manually inputted into a CSV file along with the class field representing the pipe name.

In ArcGIS Pro, the geoprocessing function “XY To Line” tool generated line features using the start and end points from the CSV file. We can then use the “Add Join” feature to join the class field according to the Start X and Start Y fields, since they are unique fields.

	OID *	Shape *	Start X	Start Y	End X	End Y	Shape_Length	Class
1	1	Polyline	-2.9813	-0.8359	-3.0175	-0.8374	0.036231	1
2	2	Polyline	-0.6233	-4.5039	-1.8963	-1.3099	3.438338	1
3	3	Polyline	-1.8423	-1.3909	-2.0443	-1.1821	0.290519	1
4	4	Polyline	-1.984	-1.2438	-2.2832	-1.107	0.328991	1
5	5	Polyline	-2.2267	-1.1415	-2.8239	-0.8841	0.65031	1
6	6	Polyline	-2.7262	-0.9177	-3.0081	-0.8246	0.296876	1
7	7	Polyline	-2.9318	-0.8521	-3.0094	-0.8226	0.083018	1
8	8	Polyline	-0.5828	-4.6476	-0.6408	-4.4952	0.163064	1
9	9	Polyline	-0.5308	-4.7727	-0.626	-4.5345	0.25652	1
10	10	Polyline	-0.5485	-4.6955	-0.5614	-4.7355	0.042029	1
11	11	Polyline	-0.546	-4.7181	-0.5541	-4.7195	0.00822	1
12	12	Polyline	-0.0249	-5.9987	-0.4684	-4.8939	1.190494	1
13	13	Polyline	-0.4148	-5.0353	-0.5177	-4.7737	0.28111	1
14	14	Polyline	-0.4753	-4.8724	-0.5681	-4.6745	0.218578	1
15	15	Polyline	-0.5346	-4.7289	-0.5522	-4.7187	0.020342	1
16	16	Polyline	-3.3773	0.0734	-3.3733	0.0598	0.014176	2
17	17	Polyline	-2.4783	0.1937	-1.262	-2.7287	3.165408	2
18	18	Polyline	-2.7192	0.3246	-3.1771	0.1475	0.490955	2
19	19	Polyline	-2.7182	0.3469	-2.4563	0.2334	0.285436	2
20	20	Polyline	-2.5792	0.2969	-2.7663	0.338	0.191561	2
21	21	Polyline	-2.7957	0.3297	-2.5914	0.3027	0.206076	2
22	22	Polyline	-2.5742	0.357	-2.4835	0.1804	0.19853	2

Figure 13: A class field was added to the attribute table, according to the class ID of the cylinders.



Figure 14: The pipes' symbology can be customized to classify by class ID.

Conclusion of the step workflow

The workflow successfully transformed a noisy, combined point cloud data into cylindrical models then into line network. A total of 43 cylinders were used in the fitting process across three pipes, representing a dramatic reduction from dense point cloud data to just 43 pairs of start/end coordinates while preserving essential geometric information.

Slicing was used to conveniently work with the polygon selection tool. Adaptive radius fitting was used for the predictable pipes, with long and straight sections and high point density, objectively determining the pipe radius, whereas an initial estimation was used for the smaller pipes, with bents and curves, and easily converged to an acceptable fit to the original data. Further corrections and smoothing were applied after merging the point clouds.

This project shows that the cylindrical fitting can be effective storing cylindrical-line point clouds efficiently, in the end the point cloud data has transformed into line data. This project also demonstrated manual and semi-supervised approach: manual slicing selection of the pipes and point sampling for the cylinder location, followed by automatic cylinder fitting algorithm performed by the software.

Task 3 – Table of coordinates of run of each pipe

Section name	Start X	Start Y	Start Z	End X	End Y	End Z
Pipe1	-2.9813	-0.8359	4.011703066	-3.0175	-0.8374	0.287140318
Pipe1	-2.9318	-0.8521	0.36234761	-3.0094	-0.8226	0.147536747
Pipe1	-2.7262	-0.9177	0.275785865	-3.0081	-0.8246	0.00786492
Pipe1	-2.2267	-1.1415	0.081076976	-2.8239	-0.8841	0.078255821
Pipe1	-1.9840	-1.2438	0.086880128	-2.2832	-1.1070	0.067317729
Pipe1	-1.8423	-1.3909	0.081095806	-2.0443	-1.1821	0.066746676
Pipe1	-0.6233	-4.5039	0.06933982	-1.8963	-1.3099	0.058591343
Pipe1	-0.5828	-4.6476	0.059427333	-0.6408	-4.4952	0.059321906
Pipe1	-0.5308	-4.7727	0.115740359	-0.6260	-4.5345	-0.12760076
Pipe1	-0.5485	-4.6955	0.020007524	-0.5614	-4.7355	-0.215008322
Pipe1	-0.5460	-4.7181	-0.164923136	-0.5541	-4.7195	-0.538801844
Pipe1	-0.5346	-4.7289	-0.500968567	-0.5522	-4.7187	-0.678709396
Pipe1	-0.4753	-4.8724	-0.56863346	-0.5681	-4.6745	-0.783831216
Pipe1	-0.4148	-5.0353	-0.736990481	-0.5177	-4.7737	-0.744591207
Pipe1	-0.0249	-5.9987	-0.745373979	-0.4684	-4.8939	-0.767507896
Pipe2	-3.3773	0.0734	4.084370032	-3.3733	0.0598	-0.323707238
Pipe2	-3.2869	0.1026	-0.718868569	-3.1105	0.1753	-0.750160619
Pipe2	-3.3383	0.0588	-0.220300077	-3.3537	0.0824	-0.791340114
Pipe2	-3.1932	0.1447	-0.587990639	-3.3848	0.0591	-0.792643669
Pipe2	-2.7957	0.3297	-0.746625386	-2.5914	0.3027	-0.749760426
Pipe2	-2.4783	0.1937	-0.745942304	-1.2620	-2.7287	-0.753217748
Pipe2	-2.5792	0.2969	-0.749338103	-2.7663	0.3380	-0.752727674
Pipe2	-2.7182	0.3469	-0.749062016	-2.4563	0.2334	-0.753441141
Pipe2	-2.5742	0.3570	-0.746838143	-2.4835	0.1804	-0.750821421
Pipe2	-2.7192	0.3246	-0.746080546	-3.1771	0.1475	-0.749648662
Pipe2	-1.3897	-2.4525	-0.740368478	-1.1804	-2.8715	-0.740908869
Pipe2	-1.2691	-2.7505	-0.698854572	-1.1572	-2.8070	-0.71610009
Pipe2	-0.9696	-2.9101	-0.743083195	-1.2435	-2.8253	-0.756103798
Pipe3	-2.8314	-1.2675	4.385783049	-2.7930	-1.3224	0.457909254
Pipe3	-2.8649	-1.2809	0.117991397	-2.5150	-1.4409	0.07166796
Pipe3	-2.8168	-1.3039	0.09773084	-2.3695	-1.5085	0.070318213
Pipe3	-2.7831	-1.3179	0.995958141	-2.8041	-1.3121	0.194850632
Pipe3	-2.7763	-1.3198	0.560451287	-2.7991	-1.3128	0.111411553
Pipe3	-2.4048	-1.4826	0.068786248	-2.3004	-1.5951	0.068151147
Pipe3	-2.3280	-1.5382	0.081574088	-2.5821	-1.4016	0.071174363
Pipe3	-2.2277	-1.8132	0.073942936	-2.2947	-1.6238	0.06776891
Pipe3	-2.2273	-1.8183	0.059349338	-1.1847	-4.5600	0.040146364
Pipe3	-1.0680	-4.8527	0.037236294	-1.2828	-4.3133	0.032868469
Pipe3	-1.0549	-4.9219	-0.123379473	-1.0418	-4.9148	-0.688603457
Pipe3	-1.0429	-4.9188	-0.015004184	-1.0443	-4.9148	-0.250777576

Pipe3	-1.0195	-4.9647	0.08367263	-1.1068	-4.7706	-0.128846394
Pipe3	-0.9554	-5.1312	-0.76180234	-0.5433	-6.1923	-0.788890457
Pipe3	-0.9937	-5.0800	-0.640452026	-1.0588	-4.9237	-0.796421649